TRNSYS

a TRaNsient SYstem Simulation program

Thermoelectric Ventilation Unit Library

Mathematical Reference







About the Manual

The information presented in this manual is intended to provide a detailed mathematical and technical reference for the Thermoelectric Ventilation Unit Library developed for TRNSYS 18. The library consists of semi-physical component models representing the thermodynamic behaviour and control of thermoelectric ventilation systems for building applications.

This manual is not intended to provide detailed reference information about the TRNSYS 18 simulation software and its utility programs. More details can be found in other parts of the TRNSYS documentation set.

Revision History

- **2021-12** — Initial implementation of the thermoelectric ventilation unit library
- **2022-08** — Updated operational modes and control strategies
- **2025-05** — Mathematical reference manual preparation

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Introduction

This manual presents a mathematical and technical reference for the *Thermoelectric Ventilation Unit Library* developed for the TRNSYS 18 simulation environment. The library was created to support detailed numerical analysis of façade-integrated decentralized air handling units (AHUs) equipped with passive and active thermoelectric heat recovery systems. The presented models were developed to enable advanced simulation of ventilation, thermal comfort, indoor air quality, and energy performance in residential buildings.

The library was developed as an extension of the modular TRNSYS framework and consists of two complementary simulation components. Type 221 represents the main semi-physical performance-map-based model of the thermoelectric ventilation unit itself, including the passive heat exchanger, active thermoelectric heat exchanger, psychrometric calculations, and performance evaluation routines. Type 230 represents the supervisory control component responsible for operational mode selection, PID-based temperature regulation, bypass management, and fan control strategies.

The numerical models were developed to support coupled building performance simulations under realistic operating conditions. The implementation enables integration with standard TRNSYS components as well as external airflow and contaminant transport solvers such as CONTAM developed by the National Institute of Standards and Technology (NIST). The coupled simulation approach allows simultaneous evaluation of thermal performance, airflow distribution, ventilation effectiveness, indoor environmental quality (IEQ), and indoor air quality (IAQ).

The thermoelectric ventilation library was designed primarily for analysis of decentralized façade-integrated AHUs intended for rapid renovation of residential buildings. Compared to conventional decentralized ventilation systems, the investigated unit incorporates both passive and active heat recovery, enabling heating and cooling of the supplied ventilation air within a compact integrated façade solution.

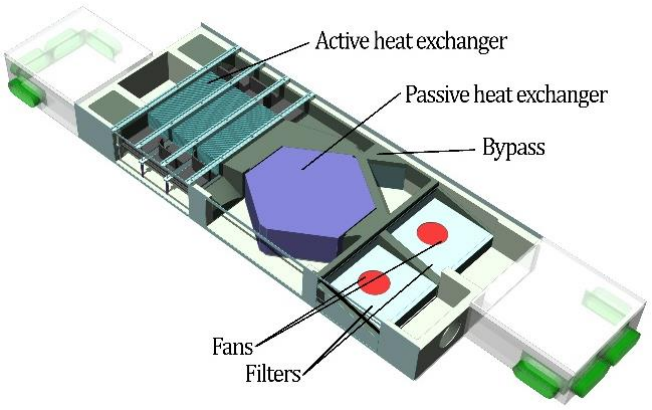


Fig. 1 **– Example design of the i**ntegrated air handling unit

Type 221 - Thermoelectric Ventilation Unit Model

Type 221 represents the principal semi-physical model of the façade-integrated thermoelectric ventilation unit. The component describes the thermodynamic behaviour of the AHU using a performance-map-based approach combining passive sensible heat recovery and active thermoelectric conditioning of the ventilation air streams.

The model consists of two interacting heat exchange stages:

- passive air-to-air sensible heat exchanger,

- active thermoelectric heat exchanger.

The passive heat exchanger is based on the constant-effectiveness minimum-capacitance method adopted from the standard TRNSYS Type 760 formulation. Sensible heat transfer between fresh and return air streams is calculated using psychrometric routines embedded within the TRNSYS environment.

The active heat exchanger consists of an array of thermoelectric cells connected in series and parallel arrangements. The thermoelectric model relies on manufacturer-derived performance maps interpolating cooling capacity and voltage as functions of electrical current and temperature difference across the thermoelectric module surfaces. Empirical thermal and electrical correction factors are further introduced to represent real operational behaviour of the integrated heat exchanger assembly.

The model calculates:

- supply and exhaust air temperatures,
- sensible heating and cooling rates,
- electrical power consumption,
- coefficient of performance (COP),
- energy efficiency ratio (EER),
- condensation risk and condensate flow rate.

Psychrometric calculations are performed iteratively across the thermoelectric cell arrangement to capture gradual temperature evolution of the air streams through the active heat exchanger.

Type 221 was developed to operate as a modular component within TRNSYS projects and can be integrated with standard building, HVAC, and control types for whole-building performance analysis. The passive heat exchanger has been adopted from the existing type 760 – Air to air heat sensible heat exchanger. This part of model uses a “constant effectiveness – minimum capacitance” approach and psychrometric calculation routine to represent the heat transfer between the two moist air streams. The sensible heat transfer between the two air streams is calculated as:

$$\dot{Q}_{pHX} = \varepsilon \cdot \min\left(C_{fr,\ amb}, C_{ret,act}\right) \cdot \left(T_{ret,aHX} - T_{fr}\right)$$

where ε represents the constant effectiveness of the heat exchanger, $T_{ret,act}$, represents return temperature from active heat exchanger, $T_{fr,amb}$ represents fresh air from ambient environment and $C_{ret,act}, C_{fr,\ amb}$ represents the heat capacities of the two air-streams, respectively. The known sensible heat transfer and inlet enthalpies given by the moist air properties are used to calculate output enthalpies of the air streams from the passive heat exchanger to the active heat exchanger or to the ambient environment

$$h_{exh} = h_{ret,aHX} - \frac{\dot{Q}_{pHX}}{\dot{m}_{ret}}.$$

$$h_{sup,pHX} = h_{fr} + \frac{\dot{Q}_{pHX}}{\dot{m}_{fr}}.$$

The **thermoelectric (TE) heat exchanger** serves as an active component in the model, providing additional control over heat transfer conditioning the supply air. The thermoelectric heat exchange calculation relies on the user-defined performance map of the selected thermoelectrical cell. The performance map is processed from the manufactures’ technical sheet provided for the selected product. The performance map provides voltage U_{TE} and cooling rate $\dot{Q}_{c,TE}$ interpolated from temperature difference between the thermoelectrical cell surfaces ΔT_{TE} and electrical current I_{TE} as controlled variable. ΔT_{TE} is given by the absolute difference between so-called hot and cold side. The theoretical voltage $U_{TE}\left(\Delta T_{TE}\ ,\ I_{TE}\right)$ and cooling rate $\dot{Q}_{c,TE}(\Delta T_{TE}\ ,\ I_{TE})$ of the thermoelectrical cell are modified by thermal

and electrical correction factors to account for real-world inefficiencies due to heat exchanger's construction and thermoelectrical cells connection configuration. The effective values of the cooling rate and voltage are corrected by the inefficiency factors *k*

$$\dot{Q}_{c,TE,eff} = \dot{Q}_{c,TE}.k_{th}$$

$$U_{TE,eff} = U_{TE} \cdot k_{el}$$

These correction factors were investigated empirically, and they are specific to the developed unit. The typical values of the thermal inefficiency factor for finned-heat exchanger types are around 0.3 to 0.5. The electrical inefficiency factor exhibits an increased voltage due to additional electrical resistance around 1.2 to 1.4.

$$\dot{Q}_{h,TE,eff} = \dot{Q}_{c,TE,eff.} + U_{TE,eff} \cdot I_{TE}$$

The interpolation and correction routine is called for each cell separately within the iteration across all cells in the series. The moist air properties of the outlet air streams from the active heat exchanger are calculated again via enthalpies within the iterative process per cells in series to calculate the gradual temperature drop or rise. The calculation uses again the embedded psychometrics routine in the TRNSYS type studio to report the outlet temperature and humidity.

$$h_{sup,aHX,i+1} = h_{sup,aHX,i} - \frac{\dot{Q}_{c,TE,eff,i\cdot}}{\dot{m}_{fr}}, with\ h_{sup,pHX}\ as\ initial\ value\ h_{sup,aHX,1}$$

$$h_{ret,aHX,i+1} = h_{ret,aHX,i} + \frac{\dot{Q}_{h,TE,eff,i\cdot}}{\dot{m}_{ret}}, with\ h_{ret}\ as\ initial\ value\ h_{ret,aHX,1}$$

The cooling and heating rate to the air streams are given by the control mode as the active heat exchanger may change the functions only by alternating the flow of the electrical current. The following equations are stated with the assumption of the cooling regime. Thus, the supply air is cooled down, while the waste heat from the opposite surface of the thermoelectrical cell is removed by the return air. In the heating regime the reversed assignment to the output heat rates are applied and the supply air is heated up.

$$\dot{Q}_{sup,aHX} = -n_{parallel} \cdot \sum_{i=1}^{n_{series}} \dot{Q}_{c,TE,eff,i\cdot}$$

$$\dot{Q}_{ret,aHX} = n_{parallel} \cdot \sum_{i=1}^{n_{series}} \dot{Q}_{h,TE,eff,i\cdot}$$

$$\dot{P}_{aHX} = n_{parallel} \cdot \sum_{i=1}^{n_{series}} U_{TE,eff,i} \cdot I_{TE}$$

The model also provides performance metrics of the active thermoelectrical heat exchanger as output. To assess the TE module efficiency, the metric COP and EER typically used for heat pump evaluation were calculated for heating and cooling regime respectively. Considering again the assumption the cooling regime, the EER is calculated at the supply air stream, while COP indicates the energy ratio on the return airstream, where the waste heat is removed and vise-verse in the heating regime.

$$COP = \frac{\dot{Q}_{ret,aHX}}{\dot{P}_{aHX}}$$

$$EER = \frac{\dot{Q}_{sup,aHX}}{\dot{P}_{aHX}}$$

The new type can also indicate the period when the condensation may occur comparing the input and outlet humidity ratio. If the difference of this ratio is greater than zero meaning that the dew point conditions were reached, the condensation risk is indicated, and condensate flow rate can be estimated using again psychrometric routine

$$\dot{m}_{cond} = \max\left(0, \quad \dot{m}_{fr} \cdot \left(\omega_{fr} - \omega_{sup}\right)\right)$$

Where $\dot{m}_{cond}$ is the estimated condenstation flow rate, $\dot{m}_{fr}$ is the mass flow rate of the air stream, ω_{fr} and ω_{sup}inlet and outlet humidity ratio to and from the AHU unit.

Model Assumptions and Limitations

The numerical model relies on several assumptions, which may limit its accuracy in specific scenarios:

- **Constant Sensible Effectiveness**: The passive heat recovery mode assumes a constant sensible effectiveness value, which may vary in real applications.

- **No Humidity Recovery**: The model does not account for latent heat recovery, focusing solely on sensible heat.

- **Empirical Performance Map for TE Module and its correction**: The TE heat exchanger performance is based on an empirical map, which could be limited in accuracy if extrapolated beyond tested conditions and for the theoretical single cell performance needs to be corrected to represent the real-life inefficiencies of the integrated cells within the heat exchanger construction as well as the electrical circuiting

- **The even airflow distribution within the cells arrangement in parallel:** The simplified even airflow distribution across the cells in parallel is assumed, neglecting the possible effect of channels surface friction and airflow profile development. The simplification is applied assuming finned heat exchanger or similar type of heat exchanger fulfilling most of the channel acting as airflow diffuser. Other type of construction may cause inaccuracies.

These assumptions simplify the model but may require further refinement or calibration for real-world applications.

Type 230 – Thermoelectric Ventilation Controller

Type 230 represents a thermoelectric ventilation controller designed to regulate air temperature and manage airflow through active and passive heat exchangers (HX) in thermoelectric conditioning of supply air. This controller dynamically modulates the electric current supplied to thermoelectric elements, controls fan operation, and adjusts the mode of the heat exchangers based on occupancy, ambient temperature, and user-specified comfort boundaries. The controller's primary function is governed by a PID control mechanism within the assembly with standard type 23 that maintains the desired supply air temperature within a specified deadband range by manipulating with the electric current in the active heat exchanger. The controller provides easy settings of multiple operational modes

Operational Modes

The model includes multiple operational modes, allowing the system to respond to various environmental conditions and user preferences:

1. **Ventilation Mode with Active and Passive HX (Mode_Active_HX = 1)**
   In this mode, the controller operates both active and passive heat exchangers to precisely regulate the supply air temperature. The PID control is used to adjust the thermoelectric current to achieve the setpoint temperature, based on the deadband range around the desired supply air temperature. This mode also incorporates bypass control, which disengages the heat exchangers when fresh air temperature is within the user-defined comfort boundaries.

2. **Discomfort Mitigation (Mode_Active_HX = 2)**
   This mode activates the heat exchangers during uncomfortable conditions outside the comfort

range. When the return air temperature exceeds the comfort boundaries, the controller adjusts the supply air temperature by activating the thermoelectric modules in either cooling or heating mode, depending on the setpoint and comfort limits.

3. **Overheating Mitigation (Mode_Active_HX = 3)**
   In this mode, the system prioritizes cooling to avoid overheating the space. The active HX is turned on if the return air temperature is above the comfort boundaries, and bypassing is enabled if the fresh air temperature lies within the comfort range. When there is no violation of the upper temperature comfort range, the ventilation mode (1) is activated

4. **Underheating Mitigation (Mode_Active_HX = 4)**
   Similar to overheating mitigation, this mode focuses on heating, engaging the thermoelectric modules when the return air temperature is below the lower comfort boundary. When there is no violation of the lower temperature comfort range, the ventilation mode (1) is activated

5. **Passive Heat Recovery Only (Mode_Active_HX = 0)**
   In this mode, only the passive HX operates, without thermoelectric conditioning. This setting is used for scenarios representing standard local unit with passive heat recovery. The supply temperature is results of the heat recovery efficiency only without any active temperature regulation.

Key Control Logic and Equations

Desired temperature control: The electric current supplied to the thermoelectric modules is proportional to the PID control signal and the rated unit electric current:

$$I_{TE} = PID\ signal\ _{Type\ 23} \cdot Rated\ Electric\ current$$

The controller uses a Proportional-Integral-Derivative (PID) control approach in assembly with standard type 23 to maintain the setpoint temperature:

- **Setpoint Temperature**: To satisfy all operational regimes, the setpoints can defined by user as desired supply temperature setpoint or by comfort temperatures boundaries with respect to selected operational mode.

- **Controlled Variable**: Monitored supply air temperature or return air temperature with respect to selected operational mode.

- **Gain and Integral Time**: Adjusted based on user-defined gain tuning and integral time tuning ratios.

Heating or cooling mode control employs a decision-making process to dynamically select either cooling or heating modes based on a comparison between the **setpoint supply air temperature** and the **heat recovery outlet temperature**. This decision is influenced by the presence of a **deadband** around the setpoint, which prevents frequent switching between modes due to minor temperature fluctuations.

1. **Cooling Mode Activation**: When the sensor heat recovery outlet temperature exceeds the supply air temperature setpoint plus the defined deadband, cooling mode is engaged. The thermoelectric cooling mode is activated (HX Mode = 0), and the PID controller adjusts the electric current to the thermoelectric elements to drive the supply air temperature toward the relevant setpoint for cooling with respect to selected with respect to selected mode

2. **Heating Mode Activation**: If the setpoint temperature exceeds the sensor heat recovery outlet temperature, the unit enters heating mode. In this mode, the thermoelectric system provides heating (HX Mode = 1), ensuring the relevant setpoint for heating with respect to selected mode

Bypass control: The bypass control disengages the heat recovery unit if fresh air temperature falls within the defined comfort boundaries, reducing unnecessary energy consumption. The following roles is applied for both passive as well as active heat exchanger:

$$Bypass\ signal = \begin{cases} 1\ if\ T_{fr} \in \left|T_{lower\ comfort}, T_{upper\ comfort}\right| \\ 0\ otherwise \end{cases}$$

Fan Control Signal in Ventilation Modes: Type 230 provides two distinct fan control mechanisms to regulate air circulation in the ventilation mode. The fan control signal adjusts the fan operation in response to real-time conditions, either based on a constant schedule or according to occupancy, which helps optimize airflow while conserving energy.

1. In **Direct Fan Control Mode (Mode_Fans = 1)**, the fan control signal directly follows a predefined schedule, maintaining a steady air circulation rate as per the Schedule_Fans parameter. This schedule defines the baseline fan speed and remains active regardless of occupancy. When the Schedule_Fans input is active, the fan operates continuously to ensure a consistent airflow through the ventilation unit, enabling precise regulation of the supply air temperature through both active and passive heat exchangers. This mode can be used for advanced control strategies e.g. CO2 based control with external logic.

2. In **Occupancy-Based Fan Control Mode (Mode_Fans = 0)**, the fan speed is modulated according to occupancy, optimizing energy use by adjusting airflow based on the number of occupants. The fan control signal in this mode is determined by both the occupancy schedule (Schedule_Occupancy) and the rated unit flow rate (Rated_Unit_Flowrate). This adaptive approach helps to reduce unnecessary energy consumption during periods of low or no occupancy. The fan control signal for occupancy-based modulation is defined as:

$$Fan\ control\ Signal\ = \min\left(\frac{Schedule\ occupancy\ \cdot 25\ \cdot 1.2}{Rated\ Unit\ Flowrate}, 1\right)$$

Where, the factor 25×1.2 represents the recommended mass flowrate per occupant in kg/h and schedule occupancy indicates number of occupants in related zone in range 0 to +∞ for given time step. To generate the fan control signal, the required mass flow rate is divided by the rated unit flow rate further adjusted ensuring that airflow never exceed design limits

In summary, Type 221 and Type 230 form a complementary system within the TRNSYS environment, accurately modelling and controlling the operation of a thermoelectric ventilation unit. Type 221 serves as the main performance-map-based model, focusing on the physical behaviour of the unit itself, while Type 230 functions as the unit's controller, responsible for regulating its operational modes and maintaining desired thermal conditions. Together, these models support a modular approach as new types within the well-established TRNSYS simulation environment, enhancing its analytical capabilities for thermoelectric ventilation applications as demonstrated in the comprehensive study presented in the current research.

TRNSYS Simulation Implementation

The thermoelectric ventilation unit library is intended to be used as a modular extension of the TRNSYS 18 simulation environment. The recommended TRNSYS implementation consists of the thermoelectric ventilation unit model, Type221, connected with the supervisory controller, Type230, and the standard PID controller, Type23. In this configuration, Type221 represents the physical behaviour of the façade-integrated AHU, while Type230 defines the operational regime, bypass state, fan control signal, and heating/cooling mode. Type23 provides the continuous control signal used by Type230 to modulate the electrical current supplied to the thermoelectric heat exchanger.

In a typical TRNSYS project, the unit is connected between the fresh-air and return-air flow paths. The fresh-air fan supplies outdoor air to the unit, while the return-air fan extracts air from the zone. Thermal loss components may be added upstream of the unit to represent heat exchange with the façade cavity, duct sections, or local installation environment. The outlet conditions calculated by Type221 are then connected to the corresponding supply-air and exhaust-air capacity or air-node components. This arrangement allows the AHU to be represented as a complete local ventilation device within a whole-building simulation.

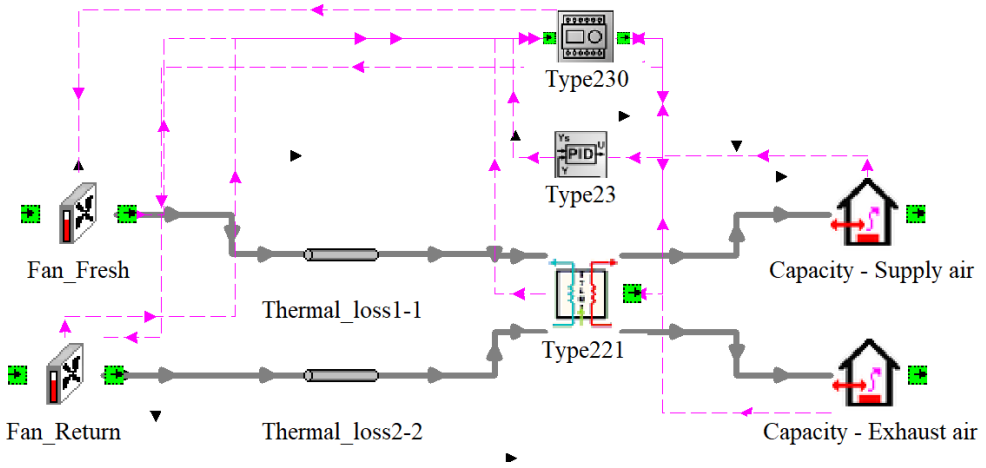


Fig. 2 - Example implementation of the thermoelectric ventilation unit library in the TRNSYS Simulation Studio environment.

The control loop is formed by feeding relevant temperature signals from Type221 and the building model into Type230. These typically include fresh-air temperature, return-air temperature, supply-air temperature, exhaust-air temperature, and heat-recovery outlet temperature. Type230 then determines whether the unit operates in passive heat recovery, active heating, active cooling, or bypass mode. The PID controller receives the setpoint and controlled variable from Type230 and returns a normalized control signal, which is converted into the thermoelectric current supplied to Type221.

An optional extension of the presented library is the coupling of TRNSYS with the CONTAM multizone airflow and contaminant transport model developed by the National Institute of Standards and Technology (NIST). CONTAM enables the simulation of interzonal airflow distribution, infiltration, exfiltration, and contaminant transport within buildings, including CO_2 concentration analysis and indoor air quality assessment.

The TRNSYS–CONTAM coupling is implemented using the NIST Type98 interface, which enables data exchange between the TRNSYS thermal model and the CONTAM airflow solver using socket-based communication. In this coupled approach, TRNSYS evaluates the thermal response and HVAC operation, while CONTAM calculates airflow distribution and contaminant transport within the building zones.

For coupled simulations, the CONTAM3DExport and Type56-98Coupler utilities may be used to automatically generate the required coupling files, including the Type98 proforma, variable exchange file, and corresponding TRNSYS building representation.

This extension is particularly suitable for demand-controlled ventilation applications. In such a configuration, CONTAM may calculate zone CO_2 concentration levels and airflow balance, while Type230 utilizes these variables as additional supervisory control inputs. Consequently, the fan control signal and operational mode of the thermoelectric ventilation unit can be dynamically adjusted according to indoor air quality requirements instead of predefined occupancy schedules only. This approach enables advanced assessment of thermal comfort, indoor air quality, and energy performance within a unified simulation framework.

Additional information regarding CONTAM, Type98, and TRNSYS–CONTAM co-simulation is available from the NIST Multizone Modeling website:

NIST CONTAM Documentation https://www.nist.gov/el/beed/nist-multizone-modeling/software/contam/contam-documentation

NIST CONTAM Software Tools https://www.nist.gov/el/energy-and-environment-division-73200/nist-multizone-modeling/software-tools

NIST Type98 Documentation https://www.nist.gov/el/beed/nist-multizone-modeling/software/type98